# Gravitational potential energy

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Relate height change to near-surface gravitational energy.

**SUCCESS CRITERIA - I can...**

1. Choose a height reference.
2. Use $\Delta U = mg\Delta h$ with the stated g.
3. Distinguish potential energy from force.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

A 3 kg object slows from 7 to 5 m/s. Find net work and check its sign.

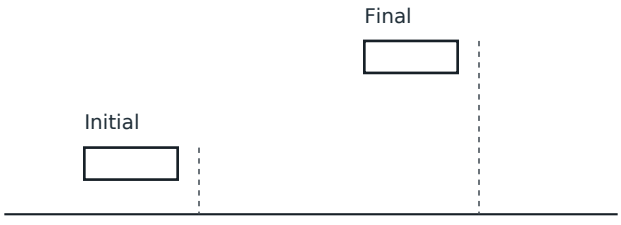
**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the model

**Reference:** $\Delta U = mg\,\Delta h = mg(h_f - h_i)$.
Approximately constant g = 10 N/kg. Choose a zero height; the energy change does not depend on that choice.

**Worked reasoning:** Raising a 2 kg object by 3 m with g=10 N/kg increases gravitational potential energy by mgh=60 J. The chosen zero does not affect this change.

**Your turn:** Draw initial and final height levels and label the 3 m difference.

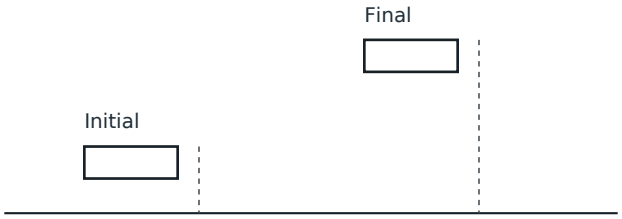



Choose and label a zero-height level.

**Marzano check - circle:** ① ② ③

## 03 Read and interpret the model

Two identical books sit at heights 1 and 3 m above the floor. Which has more gravitational potential energy relative to the floor?

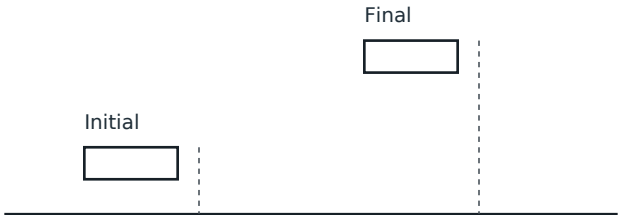



Choose and label a zero-height level.

**Marzano check - circle:** ① ② ③

## 04 Apply the physics

A 5 kg load rises from 4 m to 7 m above the floor. Use g = 10 N/kg. Find its change in gravitational potential energy. Repeat with U = 0 at a level 2 m above the floor. Label both height references and explain what stays unchanged.




Choose and label a zero-height level.

**Marzano check - circle:** ① ② ③

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Continue down each column.

## 05 Explain your reasoning

Does a negative value of U relative to a chosen zero imply a negative mass or impossible height?

**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

Two 4 kg carts at +6 and -4 m/s stick. Find final velocity and use kinetic energy to explain why this is not elastic. Use before and after snapshots to organize the momentum account.

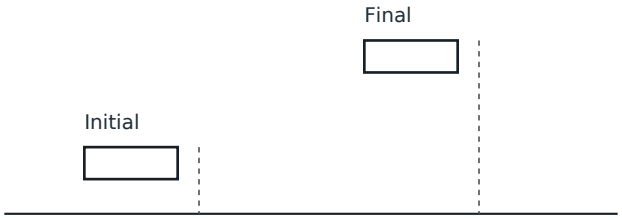
| Before | After |
| --- | --- |
| | |

Label mass, velocity and momentum.

**Marzano check - circle:** 1 2 3

## 07 Exit check

A 2 kg object descends 4 m. Find ΔU and work by gravity. Label the relevant heights or distances on the model. Use g = 10 N/kg when needed.

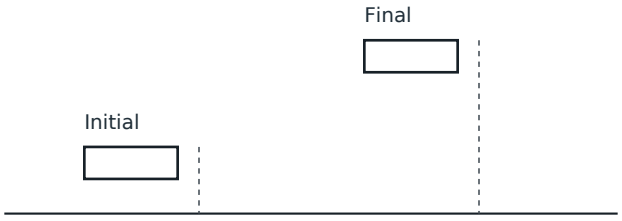



Choose and label a zero-height level.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the model

A 3 kg object moves from 2 m below a chosen zero to 5 m above it. Use g=10. Find initial U, final U and work by gravity. Label the relevant heights or distances on the model.




Choose and label a zero-height level.

**Marzano check - circle:** 1 2 3